

Contact

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Summary

- Principal-level back-end and platform engineer and technical lead with deep experience building, modernising, and maintaining large, long-lived production systems.
- A disciplined advocate for comprehensive testing (including unit, integration, and characterisation suites run against real, ephemeral databases and browsers) with the full-stack depth to deliver everything from robust data structures to rich user interfaces.
- This expertise spans decoupling monolithic platforms into modern services using Rust (tokio/axum) and AI-directed Python on Google Cloud, as well as architecting complex systems like production knowledge-graph subsystems with incremental transitive-closure engines for DAGs in SQL.
- I apply this engineering judgement to scaling AI-assisted development safely, implementing rigorous guardrails like enforced fitness functions, contract testing, and strict CI gates to intercept 'plausible-but-wrong' generated code before it hits production.
- Based in the UK, looking for a fully remote, flexible role where technical leadership and a uncompromising approach to system correctness are valued.

Technical Skills

- Languages (fluent, hand-authored): Perl, JavaScript (ES modules, statecharts, DOM), SQL (PostgreSQL & MySQL), Bash
- Languages (production): TypeScript (private-field classes, typed test suites), Rust (tokio/axum; hand-authored services)
- AI-directed delivery: Python services via spec-driven LLM orchestration—architecture, specification, review, and test governance (not hand-authored)
- AI / Agentic: LLM agent orchestration (Claude Code skills, plugins, MCP, Agent Teams), spec-driven development (IEEE 830 / EARS / Gherkin)
- Cloud: Google Cloud Platform (Cloud Run, Cloud Build, IAM, IAP, VPC, PSC, AlloyDB)
- Infrastructure: Terraform, Ansible, Docker, CI/CD pipelines, automated deployments
- Data & Streaming: PostgreSQL, MySQL, Apache Kafka, Debezium, Redis
- Frontend: XState statecharts (hierarchical, actor-model, history states), Svelte, graph visualisation (graphology/sigma), HTMX/server-rendered HTML, Rollup, monorepo workspaces
- Systems & Ops: Linux, Git, systemd
- Web: HTML/CSS, Apache HTTP Server

Employment

Situation Publishing

December 2022 to present (Tech Lead)

May 2017 to December 2022 (Senior Developer)

Lead engineer for the back-end systems powering The Register and the company's lead-generation platform, covering cloud architecture, legacy modernisation, and infrastructure automation.

- Cut call centre contact searches to near-instant by rewriting the lead-generation matching logic for a PostgreSQL-backed CRM, using denormalised, heavily indexed tables (B-tree, GIN trigram, tsvector), Redis per-campaign queues, and Kafka event logs. All backed by a comprehensive suite of unit and integration tests executed against real, ephemeral databases.
- Designed and built a keyword knowledge-graph subsystem end to end: an incremental transitive-closure engine for directed acyclic graphs in SQL (implemented from the research literature as composable ORM components); a WordNet-backed search-term pipeline with a finite-state tokeniser, double-metaphone phonetic matching, word segmentation, and Unicode (NFKC) normalisation; and an XState/Svelte/sigma front-end for visual editing of the graph.
- Progressively decoupling legacy Perl logic by building modern API endpoints, automation utilities, and event-driven Kafka and Debezium pipelines. All with thorough, multi-layered testing strategies.
- Built production Rust services (tokio/axum) to carve functionality out of the Perl monolith: a short-URL redirector with GeoIP routing, a Kafka-to-PostgreSQL lead pipeline, a database replication monitor, and a signature-verified SendGrid webhook handler. Authored Infrastructure-as-Code (IaC) Terraform for reproducible deployments.
- Directed AI agents, through my spec-driven development system, to build production Python services with clean architecture and comprehensive tests: a domain-driven CRM microservice replacing the legacy platform, and a Django/Wagtail CMS on Cloud Run/AlloyDB.
- Own the GCP estate as modular Terraform: 30+ services across Cloud Run, AlloyDB, Managed Kafka, Private Service Connect, Certificate Manager, and Cloud Build, with Packer image builds, Ansible/systemd deployment, and GitHub Actions CI/CD.
- Designing and optimising PostgreSQL/MySQL schemas and queries, and diagnosing system-wide production issues from databases to cloud services.
- Instrument services with OpenTelemetry and structured logging, and test against real, ephemeral databases and browsers (testcontainers, Playwright).
- Built an agentic development workflow where specialised LLM agents produce IEEE 830 specifications, refine them into EARS syntax and Gherkin scenarios, then validate generated code against the resulting BDD/TDD suite and bespoke LLM-oriented coding standards.

SpareRoom.co.uk

August 2011 to May 2017 (Technical Lead)

Technical Lead for one of the UK's busiest flat-sharing websites, driving improvements to code quality, performance, scalability, and development processes.

Key achievements:

- Led the migration from managed servers to colocation, successfully switching live traffic to the new infrastructure with zero downtime and zero issues. Refactored the codebase to remove years of hard-coded, hardware-specific configurations. Built a highly available, HAProxy-fronted MySQL read-write and read-only replica architecture to support the move. Used Puppet for fully reproducible server builds.
- Introduced Perl coding standards, code reviews, unit testing, and structured refactoring practices.
- Migrated version control from Subversion to Git.
- Improved scalability with memcached and Gearman; upgraded legacy Perl and MySQL systems.
- Built standardised development/testing environments using VirtualBox.
- Moved advert photo storage to AWS S3 behind EC2/ELB.
- Implemented HAProxy for front-end and database load balancing.
- Automated deployments using Perl, Rex, and Pinto.
- Led the migration of core search from MySQL to Sphinx Search, delivering major performance gains.

Self Employment

August 2005 to August 2011

Freelance web development and IT support for small businesses, including building database-driven sites (Linux, Apache/nginx, MySQL, PHP, Perl), creating custom CMS platforms, and delivering bespoke data-processing utilities.

Cheshire Building Society

July 2001 to April 2003 (Technical Support Programmer)

April 2003 to August 2005 (Infrastructure & Security Analyst)

Third-level support for critical systems, followed by infrastructure and security work covering disaster recovery testing and firewall maintenance.

Building Design Partnership

July 1996 to July 2001 (Third-Level Technical Support)

Expert Systems Design

July 1995 to July 1996 (Developer & On-Site Support)

Atkins Odlin & Partners

April 1992 to July 1995 (Computer Administrator & AutoCAD Technician)